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Research

Lake Superior Manoomin cultural and ecosystem characterization study

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ABSTRACT. Manoomin, wild rice, is integral to the culture, livelihood, and identity of the Anishinaabeg, the indigenous peoples of Canada and the United States that include the Odawa, Ojibwe, Potawatomi, and Algonquin peoples. In addition to the vital role Manoomin has in the lives of the Anishinaabeg, Manoomin is recognized as being ecologically important, feeding migrating and resident wildlife species, providing a nursery for fish and nesting and breeding habitats for many waterfowl and muskrat, and stabilizing shorelines. This study was initiated by a team of Lake Superior basin Anishinaabe communities and federal and state agencies to document and characterize (1) the importance of Manoomin habitat to Anishinaabe cultural perspectives and identity, community connections, spiritual practices, food sovereignty, and food security; and (2) the ecological importance of Manoomin habitat as an indicator of a high-quality, high-functioning, and biodiverse ecosystem. The team applied a set of cultural and ecological metrics to characterize seven case study sites around Lake Superior and used a habitat equivalency analysis to determine the amount of restoration needed to counterbalance the lost Manoomin habitat functionality. Results from this study highlight the difficulty in restoring the cultural and ecological functionality of degraded Manoomin habitat and the importance of preserving and protecting existing Manoomin habitat.

Key Words: *Cultural characterization; ecosystem services; Great Lakes; wild rice*

INTRODUCTION

Manoomin (wild rice or *Zizania palustris*) is integral to the culture, livelihood, and identity of the Anishinaabe, a group of Indigenous peoples within Canada and the United States that include the Odawa, Ojibwe, Potawatomi, and Algonquin peoples. The arrival of the Ojibwe Anishinaabe to the Great Lakes basin was in fulfillment of the prophecy that guided their migration from the Atlantic Northeast westward. Manoomin is a sacred symbol; it represents the Anishinaabe's journey as well as their relationship to the land and their identity as a culture (Minnesota Tribal Wild Rice Task Force 2018). For the Anishinaabe, Manoomin is considered a sacred, animate, more-than-human being and not an inanimate resource. It accompanies all ceremonies, celebrations, feasts, funerals, and initiations as a food source and a spiritual presence (David et al. 2019).

Manoomin grows only in the clean waters and wetlands of the Gichi-manidoo gitigaan (The Great Spirits Garden) and provides a vast array of Indigenous cultural ecosystem services that support social-ecological resilience (Mucioki et al. 2021). Although Manoomin is a natural part of the landscape, the concept of the Great Spirits Garden denotes the importance of caring for and tending to Manoomin to enhance its health and productivity, thus reflecting a reciprocal relationship (David et al. 2019). Manoomin stewardship includes spiritual and biological approaches. Tending, harvesting, and processing Manoomin was led by women prior to the Great Depression era, and women established traditional rights to a bed in many regions through the practice of binding, bunching, and tying stalks together to protect the ripening seeds (Child 2012). Spiritual practices, including ceremonies, are important to protecting Manoomin and ensuring its abundance (David et al. 2019). In addition, rice chiefs, traditionally both men and women, monitor water levels, predators, competing vegetation, and other threats and, when necessary and feasible, take actions to protect and enhance Manoomin habitat (Child 2012, David et al. 2019). Actions taken historically and today may include altering water levels, regulating impacts from beavers or blackbirds (e.g., erecting perches for predatory birds to use), removing competing vegetation, and seeding areas of potential habitat (Kinew 1995, David et al. 2019).

Manoomin is a healthy, traditional food source for the Anishinaabe that, as a major source of vitamins, minerals, fiber, and protein, helps to prevent chronic diseases (Fond du Lac Band of Lake Superior Chippewa 2018). Manoomin harvesting is integral to Anishinaabeg culture as a major community event that strengthens bonds within families and the community and provides cardiovascular benefits from the physical activity associated with harvest (Ballinger 2018, Fond du Lac Band 2018, David et al. 2019). Manoomin can be stored and consumed year-round, providing food sovereignty and a source of income for the Anishinaabe (David et al. 2019). Manoomin is so fundamental to Anishinaabe identity and culture that Anishinaabe treaties with the U.S. government guarantee access to Manoomin. The Treaties of 1837, 1842, and 1854 retain gathering rights for Manoomin (among other rights) in lands ceded to the United States. The rights to rice waters explicitly reserved in these treaties have been fundamental to Anishinaabe life historically and currently and ensure Manoomin's central place in Anishinaabe culture through

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religious, ceremonial, medicinal, subsistence, and economic uses (David et al. 2019).

In addition to Manoomin's vital role in the lives of the Anishinaabe, it is also an essential part of the Great Lakes ecosystem and environment. Natural Manoomin beds are part of complex aquatic ecosystems that support wildlife and waterfowl. Ducks, geese, swans, muskrat, deer, and moose all feed on wild rice. Additionally, insect larvae feed on Manoomin and, in turn, birds feed on these insects. Decaying Manoomin supports invertebrates that support birds, fish, and amphibians (Raster and Hill 2017, Minnesota Tribal Wild Rice Task Force 2018). Manoomin beds provide breeding and resting grounds for migratory birds, rearing habitat for resident bird species (Raster and Hill 2017), and nursery areas for young fish and amphibians (Fletcher and Christin 2015). Manoomin also plays an important role in maintaining ecosystem quality by sequestering nutrients, enriching soils, and countering nutrient loading and its negative impacts, such as algal growth and turbidity (Loew and Thannum 2011, Fletcher and Christin 2015, Minnesota Tribal Wild Rice Task Force 2018). Manoomin is also an indicator of overall water quality and ecosystem health because it is highly sensitive to changes in water quality (David et al. 2019).

Manoomin and its associated wetland habitat face many threats. It is highly sensitive to changes in water quality and requires unpolluted water to flourish; high sulfate levels (over 10 parts per million) are detrimental to Manoomin (Moyle 1944, Pastor et al. 2017, David et al. 2019, Vogt 2020). Manoomin also depends on shallow waters; both natural causes (e.g., beaver dams) and human-based causes (e.g., man-made dams and ditching) can alter lakes and rivers to make them inhospitable to this plant (David et al. 2019). Managed water-level regimes can allow other native and invasive plant species (e.g., ginoozhegoons or pickerelweed, purple loosestrife, milfoil, cattail, and pondweed) to outcompete Manoomin for habitat (David et al. 2019). In addition, climate change has already negatively affected Manoomin and it is projected to do so in the future (Stults et al. 2016, Great Lakes Indian Fish and Wildlife Commission [GLIFWC] 2018). Warmer temperatures resulting from climate change have likely reduced Manoomin abundance by favoring outcompeting plants that are better adapted for warmer climates. Warmer temperatures and humidity also make Manoomin more susceptible to brown spot disease, which destroys photosynthetic tissues and reduces seed production (Barton et al. 2013, Cozzetto et al. 2013, Grand Portage Band of Lake Superior Chippewa 2016, David et al. 2019). In addition, climate change is expected to lead to more frequent heavy rainfall events, which may lead to flooding that uproots or drowns Manoomin beds.

Given the social-ecological importance of Manoomin and the many threats Manoomin faces, a diverse group of Lake Superior basin Anishinaabe communities and federal and state agencies (Box 1), supported by Abt Associates (Abt), came together to describe the importance of Manoomin in order to foster community stewardship and education and to inform Manoomin management, protection, and policy in the Lake Superior basin and throughout the Great Lakes. Our study objectives were to document and characterize (1) the importance of Manoomin habitat to Anishinaabe cultural perspectives and identity, community connections, spiritual practices, food sovereignty, and food security; and (2) the ecological importance of Manoomin habitat as an indicator of a high-quality, high-functioning, and biodiverse ecosystem.

Box 1:

Project team

The project team consists of the following entities:

- Fond du Lac Band of Lake Superior Chippewa
- Keweenaw Bay Indian Community
- Lac du Flambeau Band of Lake Superior Chippewa
- Lac Vieux Desert Band of Lake Superior Chippewa
- Grand Portage Band of Lake Superior Chippewa
- 1854 Treaty Authority
- Great Lakes Indian Fish and Wildlife Commission
- Lake Superior National Estuarine Research Reserve
- National Oceanic and Atmospheric Administration
- University of Wisconsin Sea Grant Institute
- U.S. Bureau of Indian Affairs
- Wisconsin Department of Administration

METHODS

The project team evaluated several methodologies for characterizing the cultural and ecological importance of Manoomin and its associated habitat. We reviewed the literature and used our collective knowledge of cultural and ecological characterization methodologies to develop a list of possible methods, including:

- In-person interviews or listening sessions with tribal community members to gather qualitative information about perspectives, cultural identity, and value systems.
- A case study analysis to conduct a systematic and in-depth examination of the cultural and ecological importance of Manoomin across the Lake Superior region.
- Indigenous metrics to evaluate Indigenous priorities for cultural, social, and ecological aspects of the community that are understandable and meaningful to both Indigenous and non-Indigenous ways of thinking (Donatuto et al. 2016), including themes developed by the community (Fond du Lac Band 2018).
- An ecosystem service conceptual model to link changes caused by external stressors or interventions to Manoomin through the ecological system to socioeconomic and well-being outcomes (Olander et al. 2018).

- A social-ecological keystone concept to quantify biocultural elements of Manoomin as a keystone species (Winter et al. 2018).
- A habitat equivalency analysis (HEA) to determine the amount of restoration needed as a counterbalance for habitat that has lost cultural and ecological functionality (National Oceanic and Atmospheric Administration [NOAA] 2000, 2019).

Through discussion and consensus, we developed and applied a set of criteria to evaluate possible methods:

- Non-monetary approach to valuing Manoomin, thus avoiding putting a price tag on something that is intrinsically central to the Anishinaabe culture and existence.
- Capable of combining ecological and cultural characterization into a single analysis.
- Implementable by using mainly existing data and information (i.e., the study should not involve extensive primary data collection efforts).
- Based, at least in part, on Indigenous methodologies or research for and by Indigenous people using techniques and methods drawn from their traditions and knowledge.

Ultimately we selected an innovative application of HEA that combined (1) identifying case study sites as examples of degraded and restored Manoomin habitat; (2) refining and applying cultural and ecological metrics to characterize the degraded and restored Manoomin and its habitat at the case study sites; and (3) using HEA to determine or scale the amount of restoration needed to counterbalance cultural and ecological functionality losses of Manoomin habitat over time, caused by a variety of anthropogenic stressors.

Identifying Manoomin habitats

We identified areas across the Lake Superior region with current or former Manoomin habitat. Our goal was to identify places that experienced a decline in Manoomin over time and places where restoration actions have been undertaken by resource managers to address the decline. At each site, we aimed to understand:

- The ecological conditions at the site, such as hydrology, water quality, land use, and climatic conditions;
- The cultural and ecological importance of Manoomin at each site, including Manoomin harvest and wildlife dependence on Manoomin;
- The cause of Manoomin decline, such as changes to hydrologic conditions and water quality, invasive species, warmer temperatures, extreme events, or other threats;
- The types of restoration actions undertaken, such as management of water levels, control of invasive or competitive species, seeding efforts, and other restoration actions;
- The success or failure of those restoration actions, from both cultural and ecological perspectives; and
- The timeline of degradation and restoration actions.

We first selected two pilot case studies to test and refine the approach: Big Rice Lake and Twin Lakes (Fig. 1). Once we refined the cultural and ecological metrics and the HEA approach, as described below, we then selected five additional case studies: Perch Lake, Sand Point Sloughs, Net River Impoundment and Vermillac Lake, Hiles Millpond, and Lac Vieux Desert's Rice Bay (Fig. 1). Each Band on our project team selected a case study site, focusing on places of particular importance to their Band. Case studies could be on reservation lands, in ceded territory, or elsewhere. For each site, we gathered information about the extent and timeframe of the degradation and restoration. This resulted in a range of types of Manoomin habitat degradation and restoration approaches represented in our case studies, dispersed over a broad geographical area within the Lake Superior basin. For each site, we formed a case study team (including members of our project team and other tribal, federal, or state partners with experience managing Manoomin at the site) who assessed the Manoomin habitat degradation and restoration using the cultural and ecological metrics described below.

Refining and applying cultural and ecological metrics

We developed a set of metrics to broadly measure all aspects of community health, with health defined as a coexistence among human beings, nature and natural resources, and spiritual beings (Fig. 2). We used Donatuto et al.'s (2016) indicators of Indigenous health as a foundation, and then adjusted and added to them using Fond du Lac Band's (2018) health impact assessment themes and Winter et al.'s (2018) biocultural functional groups to ultimately develop a set of cultural and ecological metrics focused on Manoomin and the Great Lakes coastal wetlands.

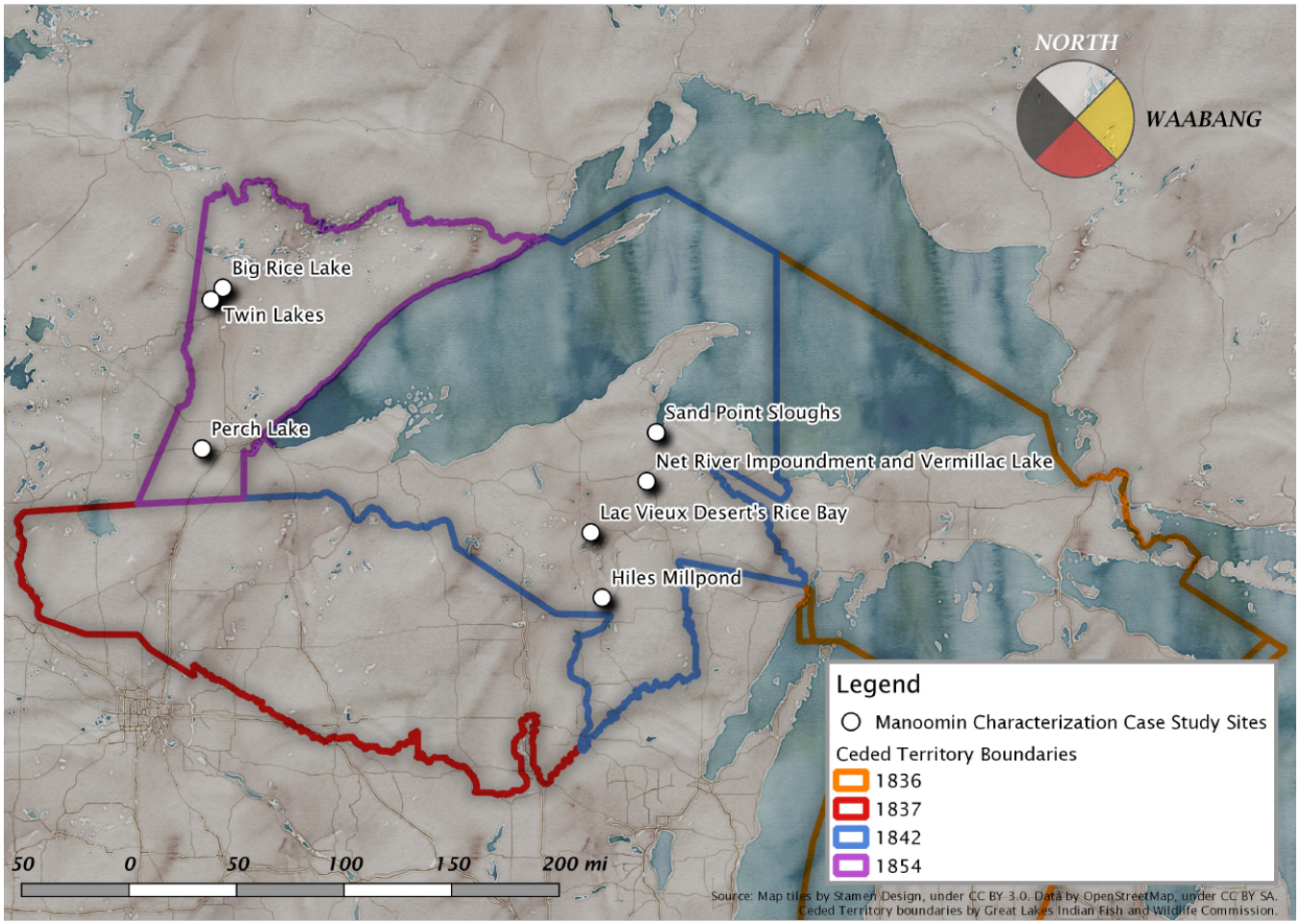
We also started with the Donatuto et al. (2016) descriptive scale and added a response choice of "No use" if tribal members did not gather, harvest, or otherwise use Manoomin at the site, or if birds and other wildlife did not feed on or use Manoomin at the site because Manoomin was of poor quality or no longer present. The descriptive scale ranks the relative status of each metric at a specific time period. We used the following five-point descriptive scale:

- We're doing great
- We're looking pretty good
- Things are not very good
- Things are very bad
- No use of Manoomin.

We left the interpretation of the descriptive scales to the respondent. We assigned percentage scores to the descriptive scales as a scalar for our HEA; our numeric scores ranged from 0% (No use) to 100% (Doing great).

We applied the draft metrics to our pilot case study during a workshop in August 2019, and refined the metrics to incorporate

Fig. 1. Map displaying the seven case study locations. The compass is in the form of a medicine wheel, an Indigenous symbol used to denote the four directions.



additional considerations, such as incorporating health into the food sovereignty metric because eating good foods relates to the health of mind, body, and spirit. Once we finalized the metrics and agreed to them on a consensus basis, we applied them to our case study sites.

For each case study site, we held a series of webinar workshops with case study teams. Overall, we held webinar discussions with 22 people on our seven case study sites, with an average of over 4 people participating per case study. During the first webinar of each case study site, the case study team identified time periods with distinct or changing Manoomin habitat conditions. This process relied on reviewing historical documents and records as well as case study team member's specific knowledge of the place. During subsequent webinars for that site, we then stepped through each time period and ranked each metric according to the scale given above. For the Anishinaabe metric, for example, we asked each case study team:

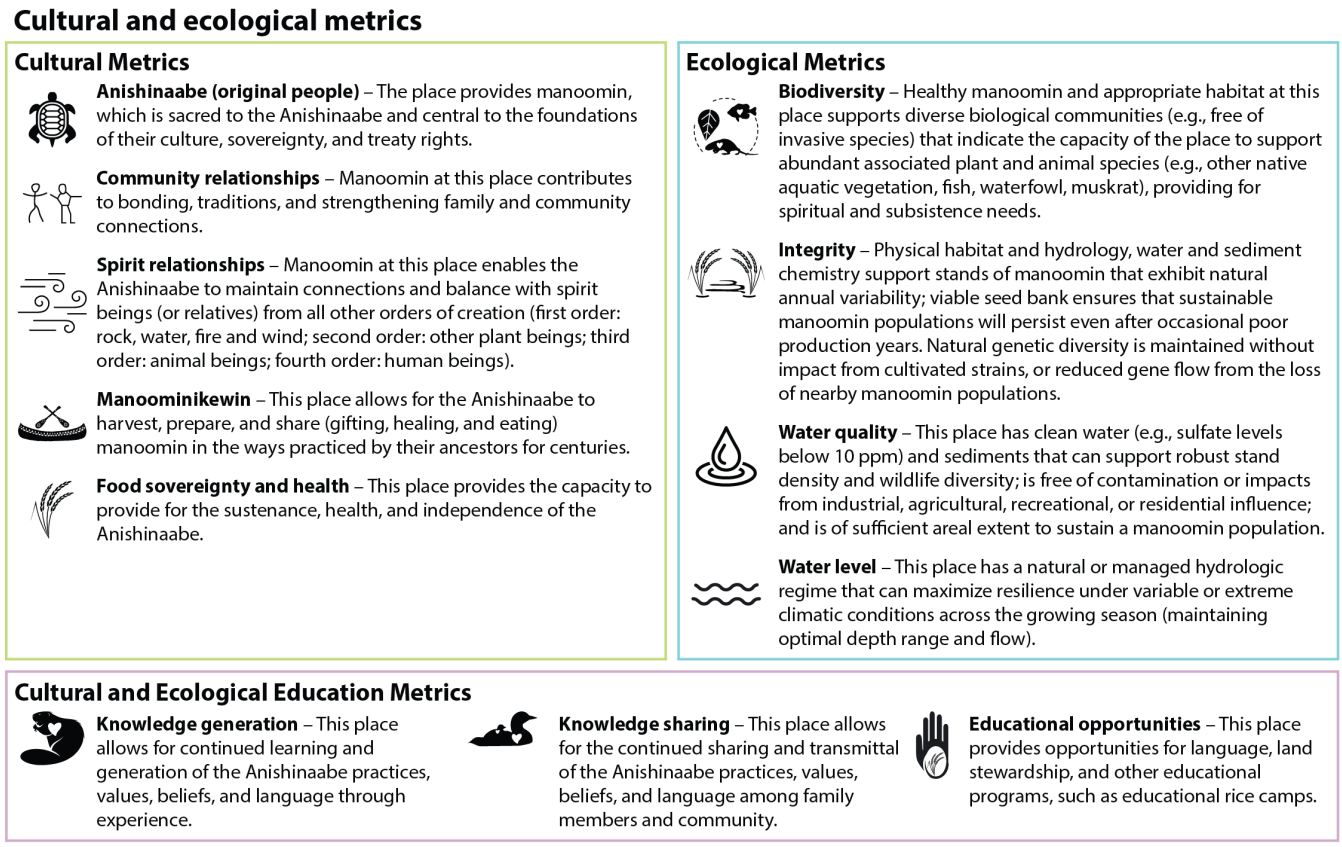
How would you rank [insert place name] in terms of providing Manoomin, which is sacred to the Anishinaabe and central to the foundations of their culture, sovereignty, and treaty rights? Would you say (a) we're doing great, (b) looking pretty good, (c) not very good, (d) things are bad, or (e) no use of Manoomin?

The case study team members individually ranked each metric, and we took an average of these rankings.

Applying HEA to characterize Manoomin

The HEA tool is commonly used by natural resource managers in natural resource damage assessments (NRDAs) to quantify a harm to a habitat, typically caused by contamination, or an ancillary harm caused during cleanup actions. In an HEA, the harm to a habitat is quantified in terms of the amount of equivalent habitat restoration needed to counterbalance the duration and magnitude of the harm (Baker et al. 2020, NOAA 2000, 2019). The restoration is referred to as "equivalent" because typically places with similar features are restored to replace the functionalities once provided by the degraded habitat. The inputs to an HEA include (1) the acreage of harmed (or degraded) habitat, (2) the severity of the degradation, (3) the timeframe over which it has occurred, and (4) an intergenerational balancing factor (or discount factor) to account for time preference, where degradation is put in present-value terms (NOAA 1999). To

Fig. 2. Cultural and ecological metrics.



quantify the acres of restoration needed to counterbalance the harm, information is also needed on the restoration, including (1) the degree of habitat improvement achieved through the restoration actions (or benefits), (2) the timeframe over which restoration benefits are achieved, and (3) an intergenerational balancing factor to account for time preference, where restoration is put in present-value terms. The output of the HEA is acres of habitat restoration.

To the Anishinaabe, particular places have cultural significance, and the standard HEA approach of replacing a particular habitat with an equivalent habitat located elsewhere is challenging. Thus, in our study, the harmed (degraded) and restored (improved) acres are the same for each case study, and we calculated the additional acreage of the same restoration that would theoretically be needed to fully counterbalance the lost functionality at the site. By using this method, it is possible that the amount of restoration in acres needed to counterbalance losses may be significantly larger than the acres of degraded habitat for many reasons, including our practical ability to restore these habitats. Manoomin restoration is challenging and, as demonstrated by the case studies, rarely achieves full functionality. If one acre of restored Manoomin wetland reaches only 50% functionality, then two acres of restored habitat are needed to counterbalance the one acre of lost Manoomin habitat. Further, habitats have typically been degraded over prolonged periods, whereas time periods of restoration benefits may be comparatively shorter, thus necessitating greater amounts of restoration to counterbalance the harm.

Case Study HEA Inputs:

- Manoomin habitat acreage: The acres of each site were characterized by the case study team. In some cases, acres included the full area of Manoomin waters; in other cases, it was only the portion of the waterbody that contained or could contain Manoomin.
- Timeframe of habitat degradation and restoration actions: On the basis of case study team members' knowledge of the site, we developed time periods for degradation (e.g., mine discharge or the installation of a dam) and restoration (e.g., reintroduction of Manoomin through seeding efforts).
- Degree of ecological and cultural habitat functionality: We then calculated the combined average of the individually ranked cultural and ecological metrics to characterize the overall degree of ecological and cultural habitat functionality for each timeframe identified at each case study site (described above). For example, if Manoomin habitat historically provided full cultural and ecological functionality, the case study team would characterize the

historic time period as "we're doing great" with around 100% cultural and ecological functionality. If a dam or other activity then decreased the cultural and ecological functionality of the Manoomin habitat, the case study team might indicate the habitat declined to "things are not very good" or "things are very bad" at 50% or 25% of functionality. On the other hand, if restoration efforts improved the cultural and ecological functionality provided by Manoomin habitat, the case study team might indicate the habitat improved to "we're looking pretty good" or around 75% of functionality.

- Intergenerational balancing factor: Because not all communities share the same time preference, we discussed the appropriate factor for this study and decided to apply a constant factor of 3% across all case studies, where past losses are compounded, and the further out in time restoration occurs, the greater the amount needed to compensate for the losses. A 3% factor is typical for ecological projects (Office of Management and Budget [OMB] 2003).

This approach allowed for a non-monetary, "equivalent acres" valuation of lost Manoomin habitat functionality. This was critical for many of the Anishinaabe communities involved in the study because it allowed for a characterization of the value of Manoomin without placing a solely monetary value on an intrinsically central aspect of their culture and existence.

RESULTS

The seven case studies, each of which profiles a story of changes in Manoomin cultural and ecological functionality over time, form the heart of this project. The case studies, grouped around the Lake Superior region, are in the 1854 Ceded Territory and the 1842 Ceded Territory (Fig. 1). Three of the seven case studies are located on reservation lands.

These case studies are primarily located in places with current or former Manoomin habitat that have experienced a decline in Manoomin over time. High water levels and stable water from dams (Lac Vieux Desert's Rice Bay), agricultural ditching (Perch Lake), and other natural and human-based causes (Big Rice Lake) have altered lakes and rivers, making them inhospitable to Manoomin. For example, lower lake levels and stable waters at Perch Lake allowed ginoozhegoons (pickerelweed) to displace Manoomin and become the dominant vegetation in the lake's rice waters. At Twin Lakes and Sand Point Sloughs, impacts from mining activity have changed the chemical composition and hydrologic conditions of these areas. At other sites, such as Hiles Millpond and Net River Impoundment and Vermillac Lake, documentation of Manoomin presence is not available from historical records; however, their physical or hydrologic features make them conducive to growing Manoomin.

Case studies also focus on restoration actions undertaken to restore Manoomin habitat over different time periods. Actions taken to improve the abundance of Manoomin included using water control structures to lower water levels (Lac Vieux Desert's Rice Bay and Hiles Millpond) or to bring lake levels to flood stage to stress ginoozhegoons, which competes with Manoomin for habitat (Perch Lake). Several case studies introduced or reintroduced Manoomin through seeding efforts (Sand Point Sloughs, Net River Impoundment and Vermillac Lake, Lac Vieux Desert's Rice Bay, and Hiles Millpond) and used intensive mechanical vegetation removal of ginoozhegoons and other competitive vegetation (Perch Lake and Big Rice Lake).

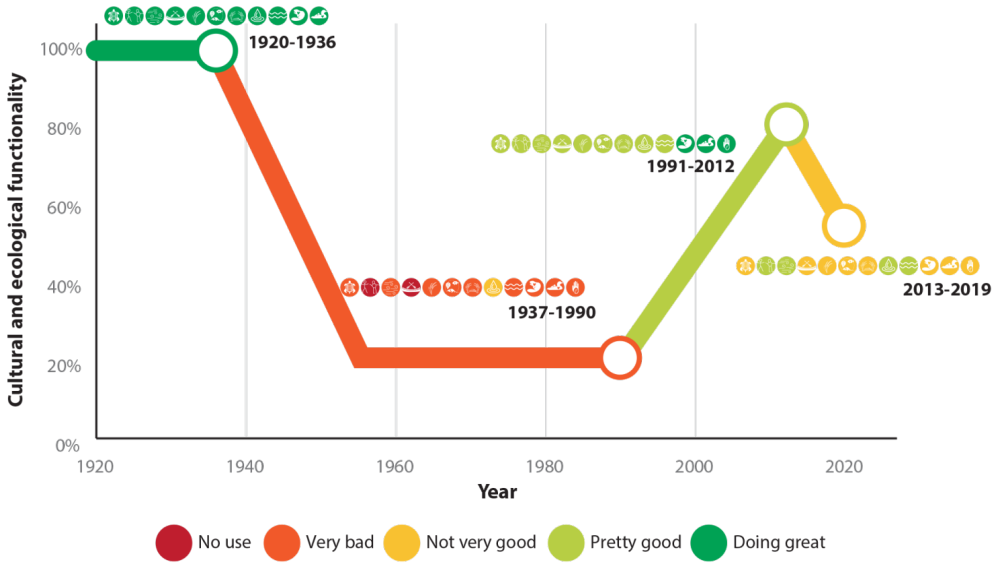
For each case study, we used our set of cultural and ecological metrics (Fig. 1) to characterize Manoomin and its associated habitat over several time periods, including periods when Manoomin habitat has declined and periods when Manoomin habitat has been restored. We then applied the HEA to demonstrate the additional equivalent units of restoration that would be needed to counterbalance the severity and timespan of degradation. For case studies where restoration actions have improved Manoomin conditions, we found that more restoration would be needed to counterbalance the lost cultural and ecological functionality. Restoration needs varied greatly. Lac Vieux Desert's Rice Bay, for example, has culturally and ecological important Manoomin beds. Although restoration efforts at Rice Bay have improved the cultural and ecological functionality provided by Manoomin since the 1990s, heavy rainfall events in more recent years have negatively affected Manoomin beds during the growing season (Fig. 3). Because of the cultural and ecological importance of Rice Bay, approximately 3034 more acres of similar Manoomin restoration are needed beyond the 243 acres of Rice Bay (or 12 equivalent restoration efforts) to counterbalance the loss that has occurred over time.

On the other hand, modest seeding and slight modifications in water-level management has successfully established Manoomin across Hiles Millpond. However, cultural use of this Millpond has been limited during much of the last century. Because of the success of restoration at Hiles Millpond and the fact that cultural and ecological functionality continue to improve (Fig. 4), only 864 more acres of similar Manoomin restoration are needed beyond the 300-acre Hiles Millpond (or nearly three equivalent restoration efforts) to counterbalance the lost habitat functionality that has occurred over time.

In some cases (e.g., Big Rice Lake and Twin Lakes), we found that recent actions have not been successful and Manoomin conditions continue to diminish. In these cases, we assumed that hypothetical actions would be taken to improve conditions in the future to demonstrate the additional equivalent units of restoration needed to counterbalance the severity and timespan of degradation. Given the history of attempting restoration with minimal response at these sites, we found that significant equivalent amounts of this restoration (e.g., tens of thousands to hundreds of thousands of acres) are needed to balance the prolonged period of degradation. These case studies demonstrate the difficulty in restoring Manoomin and its associated habitat and the importance of protecting and preserving existing Manoomin habitat to ensure a future with Manoomin.

Table 1 provides a brief overview of the case studies, including the key threats to Manoomin at these places, some of the actions taken to improve Manoomin habitat, and the HEA results that indicate how many acres of similar Manoomin restoration habitat are needed to balance lost habitat functionality over time.

Fig. 3. Characterization of cultural and ecological functionality provided by Manoomin and its associated habitat at Rice Bay.



DISCUSSION

Although each case study is unique with distinct attributes, several common themes emerged across the sites.

Restoring Manoomin and its associated habitat remain a significant challenge under current conditions.

Success of these restoration actions has been incremental and at times challenging. Restoration actions taken at historically high-producing Manoomin waters (including Big Rice Lake, Twin Lakes, Lac Vieux Desert's Rice Bay, and Perch Lake) have not returned Manoomin and its associated habitat to its full historical extent or its full cultural and ecological functionality. In some cases, restoration actions have been largely ineffective, with Manoomin abundance and density continuing to decline. For example, natural resource managers have tried to improve the conditions of Manoomin and its associated habitat at Big Rice Lake since the 1990s; however, actions have had limited success and Manoomin conditions continue to diminish. In some cases, the cultural function of Manoomin may be impaired by influences outside the scope this study.

Several case studies also highlight the need to return to the concept of traditional stewardship and to careful tending of Manoomin through sustained, long-term resource management. At Perch Lake, the Fond du Lac Band of Lake Superior Chippewa developed a management strategy that brings lake levels to flood stage every four years to stress perennial species, such as ginoozhegoons, which otherwise outcompete Manoomin. This long-term restoration approach provides Manoomin with a competitive advantage in the immediate years following the flood stage (Fond du Lac Band 2018).

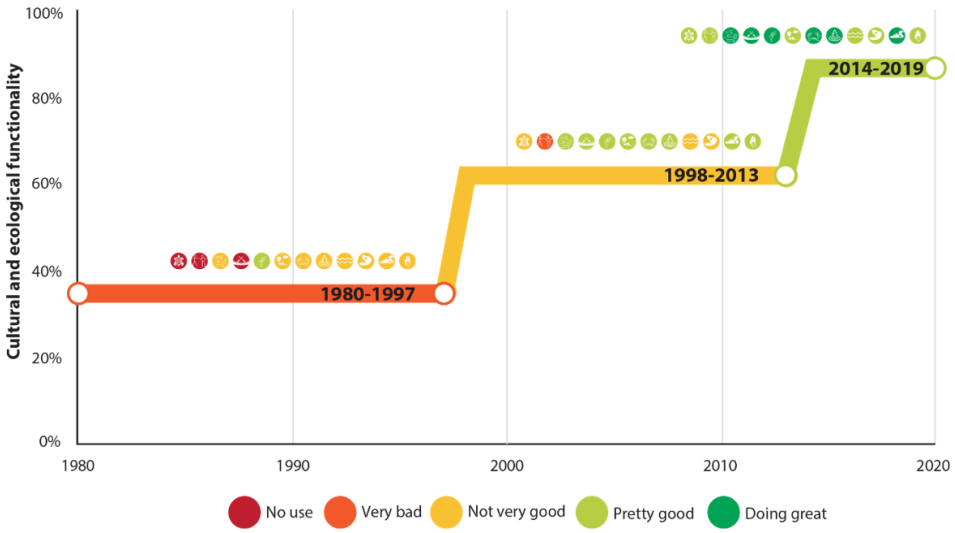
Even in places where Manoomin restoration has shown success, more restoration is often needed given the significant historical losses in Manoomin cultural and ecological functionality.

The HEA approach applied in this study accounts for the amount of time that Manoomin habitat has been degraded and the time required for restored Manoomin habitat to recover or reach improved functionality. In several case studies, water level modifications led to a decline in Manoomin habitat over 100 years ago. For example, Lac Vieux Desert was first dammed around 1870 for logging operations, and by 1907 the Wisconsin Valley Improvement Company (WVIC) began operating the lake as a storage reservoir. Changes in water levels caused by the dam initiated a decline in Manoomin, and from 1938 to 1952 Manoomin declined steadily. Community members stopped harvesting it during this period (LaBine 2017, Barton 2018). In addition, mine tailings and associated high concentrations of heavy metals have had an impact on the area and nearby Sand Point Sloughs, a culturally important site.

Even with successful restoration, Manoomin habitat at many of our case study sites has had significant cultural and ecological losses over a long period of time, which often means that many more acres of restoration are needed to counterbalance the lost habitat functionality than the case study footprint. At Lac Vieux Desert's Rice Bay, the equivalent of 12 restoration efforts (from 1991 to 2019) are needed to counterbalance the lost cultural and ecological habitat functionality (from 1900 to 1990), whereas at Sand Point Sloughs, 22 equivalent restoration efforts (from 1991 to 2019) are needed to counterbalance losses from 1920 to 1990.

At some locations, restoration actions may never fully recover all cultural and ecological functionality given that long time period of loss. At Twin Lakes, for example, actions taken to improve Manoomin and its associated habitat have been limited and have not addressed the fundamental problem of sulfate loading from the mine (Minnesota Tribal Wild Rice Task Force 2018). Given the significant cultural and ecological losses that have occurred since the installation of a tailings basin for U.S. Steel's Minntac facility in the mid-1960s, it is challenging to foresee a scenario where restoration actions could fully recover all lost functionality. In these cases, protection and/or restoration of Manoomin habitat at additional locations may be one approach to compensate for all of the losses that occurred over time.

Fig. 4. Characterization of cultural and ecological functionality provided by Manoomin and its associated habitat at Hiles Millpond.



Seeding to enhance existing Manoomin stands and to introduce it to new locations can be worthwhile and necessary; places with favorable habitat features and conditions seem conducive to growing Manoomin.

Manoomin seeding in waters with favorable physical or hydrologic features can be an effective and inexpensive way to restore Manoomin (David et al. 2019). In addition, seeding both at sites where Manoomin is known to have historically occurred and at sites where there are no records but hydrologic conditions seem suitable can be worthwhile and necessary: "worthwhile because of the many ecological and cultural benefits rice provides and because rice abundance in the state remains lower than it was prior to European contact, and necessary because rice seed has a very limited natural ability to disperse" (David et al. 2019:68). Natural resource managers around the Lake Superior region have had some success in identifying good Manoomin habitat on the basis of physical or hydrologic features, seeding Manoomin, and using lakebed sediment core samples to identify historic presence of Manoomin seed (Myrbo et al. 2011).

In two of our seven case studies, natural resource managers selected areas that were not known to have any Manoomin but were thought to have favorable conditions for Manoomin growth: suitable soils, clean water, and modifications in water-level management. Two case studies are showing preliminary success in their seeding efforts. At Hiles Millpond, biologists realized that the social-ecological benefits of this place could be significantly enhanced by establishing Manoomin. With modest seeding and slight modifications in water-level management, resource managers successfully established Manoomin across Hiles Millpond. At Net River Impoundment and Vermillac Lake, the Keweenaw Bay Indian Community (KBIC) worked with the Michigan Department of Natural Resources to identify areas for Manoomin restoration, and the Net River Impoundment and Vermillac Lake were selected as lakes with potential for Manoomin beds. After successfully seeding test plots at both lakes, KBIC expanded seeding efforts and has seen successful establishment of Manoomin at both sites. Cultural teachings and practices related to Manoomin are beginning to occur at the Net River Impoundment.

Although the results of seeding efforts are encouraging, more study is needed to confirm whether seeding can lead to culturally and ecologically high-quality Manoomin habitat. In addition, given that the period of degradation is often longer than the period of restoration, additional Manoomin restoration may be needed to counterbalance the lost habitat functionality that has occurred over time.

Restoration must be adaptive; what may have worked in the past may not be successful in the future, given additional threats.

Many tribal, state, and federal agencies have been involved in Manoomin restoration around the Lake Superior region for decades. In the case of tribal communities, the longstanding practices of caretaking Manoomin rest on a foundation of traditional ecological knowledge. However, in some cases, actions taken in the past that have had some success at restoring Manoomin are no longer successful. For example, more frequent heavy rainfall events in the spring and summer have negatively affected Manoomin in Lac Vieux Desert's Rice Bay. Above-average precipitation events, which may lead to "ghost rice" (empty seed hulls that never fill) and brown spot disease, are likely driving the decline of Manoomin abundance on Rice Bay. In addition, Sand Point Sloughs is connected to Lake Superior and is affected by more extreme oscillations in the lake's water level. These regional threats to the sloughs may be affecting Manoomin abundance and are largely beyond local control. The decrease in social-ecological functionality provided by Manoomin in recent years at several of our case study sites suggests the need for adaptive management of Manoomin habitats. Actions may need to be adjusted to respond to additional threats, such as climate change, to be successful in the future.

Table 1. Case study summaries.

Case study	Threats to Manoomin	Restoration actions to improve Manoomin	Additional restoration needed
Lac Vieux Desert's Rice Bay (characterization focused on 243 restoration acres)	High water levels caused by a concrete and steel dam at the outlet of the lake in the 1930s and above-average precipitation in the 2010s	Water level management and Manoomin seeding	3034 acres of similar Manoomin restoration needed to balance the lost habitat functionality over time, or 12 equivalent restoration efforts.
Perch Lake (characterization focused on 400 restoration acres)	High water levels caused by agricultural ditching in the 1920s; competitive vegetation caused by a non-functional dam in the 1960s	Water level management and removal of competitive vegetation	5204 acres of similar Manoomin restoration needed to balance the lost habitat functionality over time, or 13 equivalent restoration efforts.
Sand Point Sloughs (characterization focused on 8 restoration acres)	Deposited mine tailings from a copper ore processing plant that operated north of the sloughs in the 1920s; high water levels and invasive species after 2005	Manoomin seeding and remediation efforts to stabilize the tailings	175 acres of similar Manoomin restoration needed to balance the lost habitat functionality over time, or 22 equivalent restoration efforts.
Net River Impoundment and Vermillac Lake (characterization focused on 97 restoration acres)	Unclear if Manoomin historically grew at site; if it did, land use changes likely responsible for Manoomin's depletion	Manoomin seeding	1129 acres of similar Manoomin restoration needed to balance the lost habitat functionality over time, or nearly 12 equivalent restoration efforts.
Hiles Millpond (characterization focused on 300 restoration acres)	Unclear if Manoomin historically grew at site; if it did, high water levels caused by dam construction likely responsible for Manoomin's depletion	Water level management and Manoomin seeding	864 acres of similar Manoomin restoration needed to balance the lost habitat functionality over time, or 3 equivalent restoration efforts.
Big Rice Lake (characterization focused on 1870 restoration acres)	Hydrological changes and competing vegetation	Water level management and removal of competitive vegetation	Varies depending on hypothetical improvement scenario.
Twin Lakes (characterization focused on 210 acres)	Discharge of mine tailings from an iron ore processing plant upstream of the lakes since the 1960s, which has increased sulfate levels and increased water volume	Seepage collection system to collect some of the mine tailings discharge and limited Manoomin seeding and water level management	Varies depending on hypothetical improvement scenario.

Monitoring should be incorporated into all future restoration projects.

Monitoring can help wild rice chiefs and other natural resource managers assess the health of existing Manoomin habitats, evaluate the success of different restoration actions, and make informed resource management decisions. Monitoring can provide information about ecological trends, including Manoomin productivity and biomass, as well as information about other components of Manoomin waters, such as water quality and the use of waters by muskrats, beaver, geese, swans, and other beings. It can also provide information about cultural trends, such as harvest levels by tribal members and the exercise of treaty-reserved harvesting rights. Because of high variability in the productivity and biomass of Manoomin from year to year, monitoring is most useful when undertaken over several years (Kjerland 2015a). Monitoring should be completed by using methods that are both scientifically robust and culturally respectful (Kjerland 2015a, 2015b). The seven case studies in this project would not have been possible if not for existing monitoring data.

Traditional ecological knowledge can help in understanding habitat functionality across the Lake Superior region.

Cultural leaders, community members, wild rice chiefs, Manoomin harvesters, and elders have essential knowledge and perspectives that inform the characterization of the social-ecological functionality provided by Manoomin over long time periods. Many such cultural leaders shaped the development of our metrics and informed the characterization of Manoomin at specific sites. In a few instances, our project team relied on wild rice chiefs and elders to provide cultural and traditional ecological knowledge about a place. For example, the Fond du Lac Band of Lake Superior Chippewa case study team received input from an elder and wild rice chief to characterize a historic time period for Perch Lake where the case study team had limited ecological monitoring data.

Educating tribal and nontribal community members can support successful Manoomin restoration.

Although Manoomin is a social-ecological keystone plant in the Lake Superior region, the benefits and values of Manoomin are often unknown or underappreciated by the general public (David et al. 2019). Education and information about the importance of Manoomin can encourage stewardship of Manoomin and improve restoration outcomes. In one Minnesota study, 76% of ricers supported expanding Manoomin education and outreach programs (Davenport et al. 2020). On Lac Vieux Desert, for example, some lakeshore owners and boaters viewed Manoomin as a nuisance. After taking the time to educate the non-tribal community about the importance of Manoomin and why it is worth protecting, the Lac Vieux Desert Band now works closely with them to ensure the existence of Manoomin in Rice Bay.

Preserving existing Manoomin habitat is critical to ensuring a future with Manoomin.

Given the significant challenges in restoring Manoomin that has become degraded, a key management strategy for Manoomin is to protect and preserve existing Manoomin stands and the clean water resources and habitats in which they thrive. In many places, dramatic changes to wetland and lake systems, including hydrologic changes from dams, agricultural ditching, and mining activities, have unforeseen consequences. Protecting areas with Manoomin habitat could mitigate some stressors and allow the

plant to adapt to climate change and other changing conditions. Manoomin habitats may be protected through several actions, including first ensuring there is a comprehensive characterization (mapping) of the habitat across the Great Lakes region, for instance by using hyperspectral imaging to delineate Manoomin habitat. Acquisitions and conservation easements may also be part of the strategy. Instituting best management practices to protect high-quality habitat from existing stressors should also be considered. This may include controlling invasive species; limiting activities with adverse consequences, such as discharging mine waste, in sensitive habitats; and developing climate monitoring and adaptive management plans.

CONCLUSION

This study employed a blend of methodologies unique to the circumstances of Manoomin and Anishinaabeg in the Great Lakes region but is also grounded in prior research related to cultural ecosystem services, the social-ecological keystone concept as applied to a plant species, and the practice of habitat equivalency analysis that is often employed in assessing damages to natural resources (Millennium Ecosystem Assessment 2005, Winter et al. 2018, Pavanelli and Voulvoulis 2019). By drawing on these practices, the group collaboratively developed a novel and culturally appropriate approach to describe the range of gifts provisioned by a plant central to Anishinaabe life. In the development of metrics, traditional ecological knowledge was foundational to describing the full range of benefits and relationships experienced through the practices of tending and harvesting wild rice beds. To sustain and reliably harvest Manoomin over centuries, adaptive and relational management practices were developed in situ by Anishinaabe who have stewarded Manoomin and associated wetland habitat continuously through to the present. Traditional ecological knowledge held by members of the case study teams included this observational and adaptive response to changing conditions through relationship with land, water, and other living beings (Berkes et al. 2000).

The novel approach developed in this study might be applied in other circumstances where an ecological loss induces a socio-cultural loss. For example, NRDA is the process of quantifying the harm from contamination to natural resources and determining how much restoration is needed to compensate for the damage and make the public whole. This approach could be applied to characterizing and estimating damages for cultural losses in Tribal NRDAs. Because this approach assesses a loss of cultural ecosystem services using non-monetary values, it may be a more appealing methodology to many communities wishing to understand the total impact of these loses without putting a monetary value on those services. Similarly, this approach may be useful for assessing the dual cultural and ecological losses associated with climate change impacts for Indigenous people and local communities. An evaluation of social-ecological loss may enable these communities to describe potential climate resilience actions that could restore cultural ecosystem services and identify or seek adequate restitution when possible. Using this methodology, climate change resilience actions, as well as remediation and restoration, could be prioritized on the basis of their relative ability to restore the vital cultural ecosystem services that generate Indigenous knowledge, cultural identity, and social ecological resilience.

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Data Availability:

The data that support the findings of this study are available on request from the corresponding author, H. Hosterman. The case studies are publicly available at https://www.abtassociates.com/insights/publications/report/lake-superior-manoomin-cultural-and-ecosystem-characterization-study. The underlying data are not publicly available because some of the information is considered culturally sensitive, but may be available in a culturally sensitive manner upon request.

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